HKN ECE 330 Review Session

Exam 1

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Topics to Review

Ch. 2

- Single phase power
  - Real/Reactive/Complex
  - Power Factor (and Correction)
- Three phase power
  - Wye vs Delta Connection
  - Per phase Equivalence

Ch. 4

- Transformers
- Magnetic circuits
- Flux Linkage
- Coupled Coils

Past Equation Sheet

FA24

USEFUL INFORMATION

sin(x)=cos(x-90°)
0<θ<180° (lag)
-180°<θ<0 (lead)

$\bar{V} = \bar{Z}\bar{I}$ $\quad \bar{S} = \bar{V}\bar{I}^* = P + jQ$ $\quad \bar{S}_{3\varphi} = \sqrt{3}V_L I_L \angle\theta$

$I_L = \sqrt{3}I_\varphi$ (delta) $\quad \bar{Z}_Y = \bar{Z}_\Delta/3$

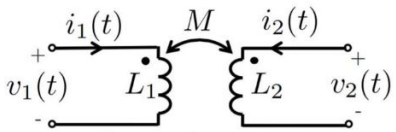
$V_L = \sqrt{3}V_\varphi$ (wye) $\quad \mu_0 = 4\pi \times 10^{-7}$H/m

ABC phase sequence has A at 0, B at -120°, and C at +120°

$\int \underline{H} \cdot \underline{dl} = \int \underline{J_f} \cdot \hat{n} dA$ $\quad \int \underline{E} \cdot \underline{dl} = -\frac{d}{dt}\left(\int \underline{B} \cdot \hat{n} dA\right)$ $\quad \mathcal{R} = \frac{l}{\mu A}$ $\quad Ni = \mathcal{R}\varphi$

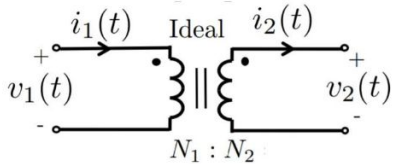
$\varphi = BA$ $\quad \lambda = N\varphi = Li$ (if linear) $\quad v = \frac{d\lambda}{dt}$ $\quad k = \frac{M}{\sqrt{L_1 L_2}}$

1hp=746 W



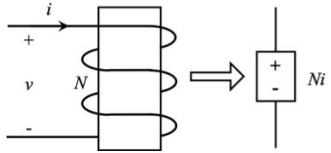
$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$

$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$



$a = \frac{N_1}{N_2}$ $\quad N_1 i_1 = N_2 i_2$

$\frac{V_1}{V_2} = \frac{N_1}{N_2}$



Single Phase Power

Average Power - AC Circuits

- For AC circuits, voltage, current, and instantaneous power have the form

$$v(t) = V_m \cos(\omega t + \theta_v)$$

$$i(t) = I_m \cos(\omega t + \theta_i)$$

$$p(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

- Average power is the average of the instantaneous power over one period:

$$P = \frac{1}{T} \int_T p(t) dt$$

$$P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i)$$

RMS Values

- In power systems, we use root mean square (RMS) values
- The RMS value of any periodic function is defined as

$$V_{rms} = \sqrt{\frac{1}{T}\int_T [v(t)]^2\,dt}$$

- For sinusoids, the RMS value is just the peak value divided by $\sqrt{2}$

$$V_{rms} = \frac{V_m}{\sqrt{2}} \qquad I_{rms} = \frac{I_m}{\sqrt{2}}$$

- Average power, using RMS values:

$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$$

Phasors Review

- Voltage and current are commonly written as phasors, using RMS values as magnitudes
- As a sinusoid:

$$v(t) = 3\sqrt{2}\cos(\omega t + 30^\circ)\ [V]$$

- As a phasor:

$$\overline{V} = 3\angle 30^\circ\ [V]$$

Real Power

- We can also find average / real power using phasors

$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$$

$$P = \mathrm{Re}\{V_{rms} I_{rms} e^{j(\theta_v - \theta_i)}\}$$

$$P = \mathrm{Re}\{\overline{V}\,\overline{I}^{*}\}$$

- Real power is what does useful work
- Measured in watts (W)

Reactive Power

- Reactive components consume/supply instantaneous power, but zero average power
- The amplitude of the instantaneous power consumed by reactive components is the reactive power

$$Q = V_{rms} I_{rms} \sin(\theta_v - \theta_i)$$

$$Q = \mathrm{Im}\{V_{rms} I_{rms} e^{j(\theta_v - \theta_i)}\}$$

$$Q = \mathrm{Im}\{\overline{V}\,\overline{I}^{*}\}$$

- Reactive power is measured in volt-amperes reactive (VARs)

Complex Power

- Real and reactive power are the real and imaginary parts of a complex quantity, called complex power

$$\overline{S} = \overline{V}\,\overline{I}^{*} = P + jQ$$

- The magnitude of complex power is called apparent power

$$S = |\overline{S}| = \sqrt{P^2 + Q^2} = V_{rms} I_{rms}$$

- Apparent power is measured in volt-amperes (VA)

Power Factor

- Average/real power:

$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$$

- Power factor angle:

$$\theta = \theta_v - \theta_i$$

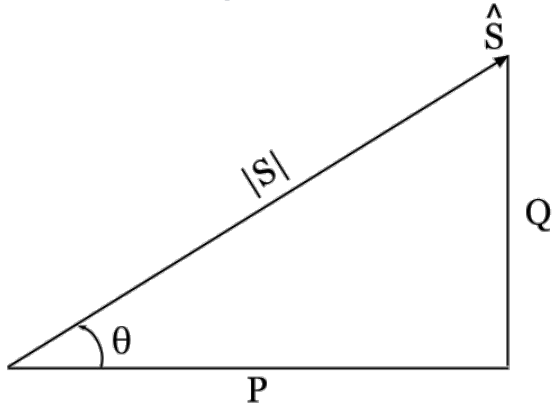
- Power factor:

$$\text{PF} = \cos(\theta) = \frac{P}{S}$$

- Power factor is the ratio of real power to apparent power

Power Factor

- Power factor is described as lagging or leading depending on if current lags or leads voltage
- **Inductive load**: current lags voltage, lagging PF (positive theta)
- **Capacitive load**: current leads voltage, leading PF (negative theta)
- **Purely resistive load**: current and voltage are in phase, unity PF



Power Factor Correction

- Ideally, power factor is close to unity
- If too much reactive power is being consumed, need capacitor bank

$$Q_{\text{add}} = Q_{\text{new}} - Q_{\text{old}}$$

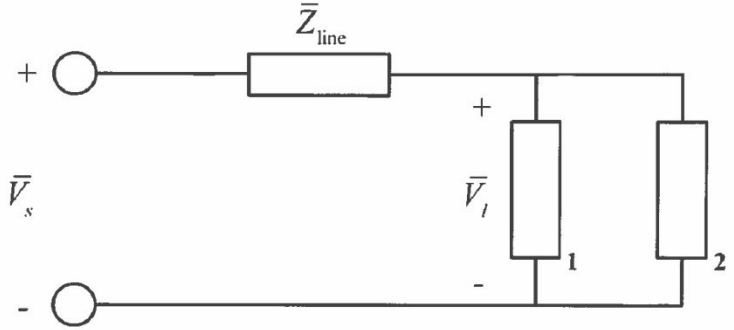
- Q_{old} often given

$$Q_{\text{new}} = P\tan(\theta_{\text{new}})$$

- Result is always **“add $|Q_{add}|$ of capacitance”**

Example: Fall 2018 Exam 1

Problem 1:



Two single-phase loads are connected in parallel to a voltage source through a feeder with impedance $\bar{Z}_{\text{line}} = 1 + j\sqrt{3}\ \Omega$. Load 1 consumes 1500 W of power at a power factor of 0.8 lagging. Load 2 consumes 1000 VA of power at a power factor of 0.6 lagging. The voltage at the loads is $v_l(t) = \sqrt{2}(120)\cos(377t)\text{V}$.

Using the given values, find:

a) The total complex power consumed by both loads.
b) The total current supplied by the source.
c) The voltage as a function of time supplied by the source $v_s(t)$.
d) If a capacitor is connected in parallel with the two loads, what power must be supplied to achieve a total power factor of 0.9 lagging?

Potentially useful formulas:

sin(x)=cos(x-90°)

$0<\theta<180°$ (lag)

$-180°<\theta<0$ (lead)

$\bar{V} = \bar{Z}\bar{I}$

$I_L = \sqrt{3}I_\varphi$ (delta)

$V_L = \sqrt{3}V_\varphi$ (wye)

$\bar{S} = \bar{V}\bar{I}^* = P + jQ$

$\bar{Z}_Y = \bar{Z}_\Delta/3$

$\mu_0 = 4\pi \times 10^{-7}\text{H/m}$

$\bar{S}_{3\varphi} = \sqrt{3}V_L I_L \angle\theta$

Example: Spring 2024 Exam 1

A single-phase source serves two loads connected in parallel.

- Load 1 draws 180 W at a lagging PF of 0.6.
- Load 2 draws 1 A at a leading PF of 0.6.

The voltage across the loads is $V = 100\angle 30°$ V. (Note that the voltage angle is not zero.)

a) Compute the current phasor I_1 for Load 1. Answer with a complex number in either polar or rectangular form

b) Compute the current phasor I_2 for Load 2. Answer with a complex number in either polar or rectangular form.

c) How many VARs of capacitance do we need to make the overall PF unity?

d) How many VARs of capacitance do we need to make the overall PF 0.8 leading?

Potentially useful formulas:

$\sin(x)=\cos(x-90°)$

$0<\theta<180°$ (lag)

$-180°<\theta<0$ (lead)

$\bar{V} = \bar{Z}\bar{I}$

$I_L = \sqrt{3} I_\varphi$ (delta)

$V_L = \sqrt{3} V_\varphi$ (wye)

$\bar{S} = \bar{V}\bar{I}^* = P + jQ$

$\bar{Z}_Y = \bar{Z}_\Delta / 3$

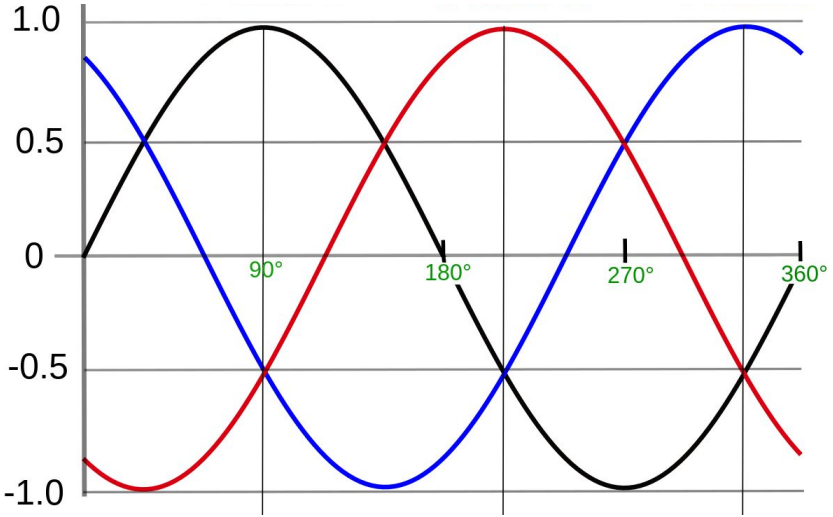
$\mu_0 = 4\pi \times 10^{-7}$ H/m

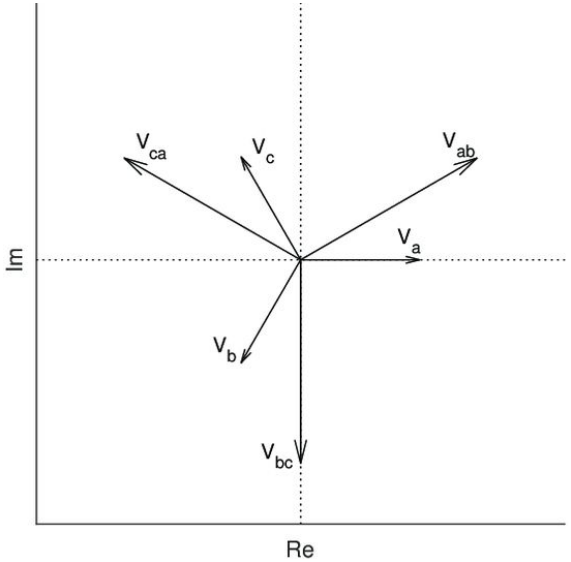
$\bar{S}_{3\varphi} = \sqrt{3} V_L I_L \angle\theta$

Three Phase Power

Three Phase Power

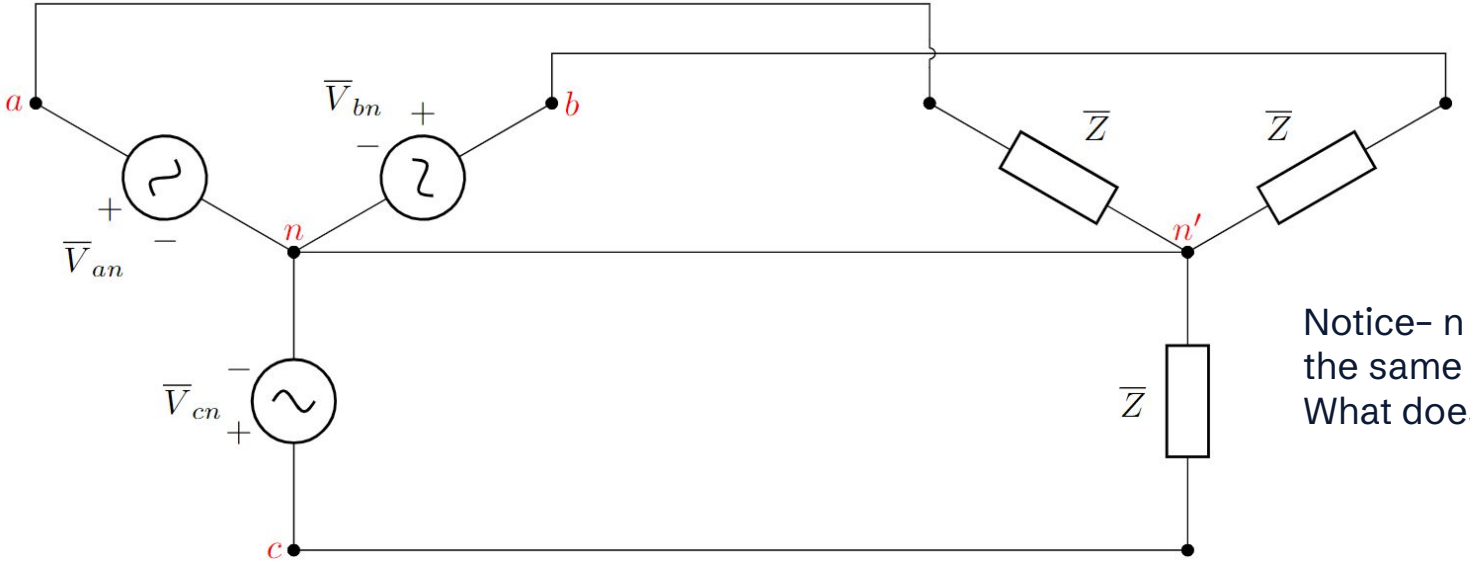
- Allows for a constant power flow
- Each phase current and voltage is separated by 120°
- Three phase power is three times the per phase power





Wye-Wye Connection

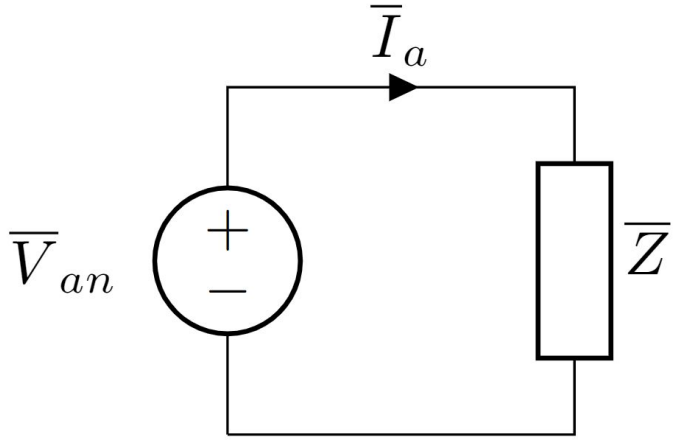
- Wye connection: line current is the same as phase current
- Line to line voltage and phase voltage are related by $V_L = \sqrt{3}V_\varphi$
- Can we simplify this? I don't want to write so many KCL/KVL equations!



Notice– n and n' are at the same potential! What does this mean?

Wye-Wye Single Phase Equivalent Circuit

- Neutral terminals of source and load have the same potential
- Analyzing a three phase wye-wye connection is like analyzing three separate single phase circuits!
- Line Current is the SAME as phase current!



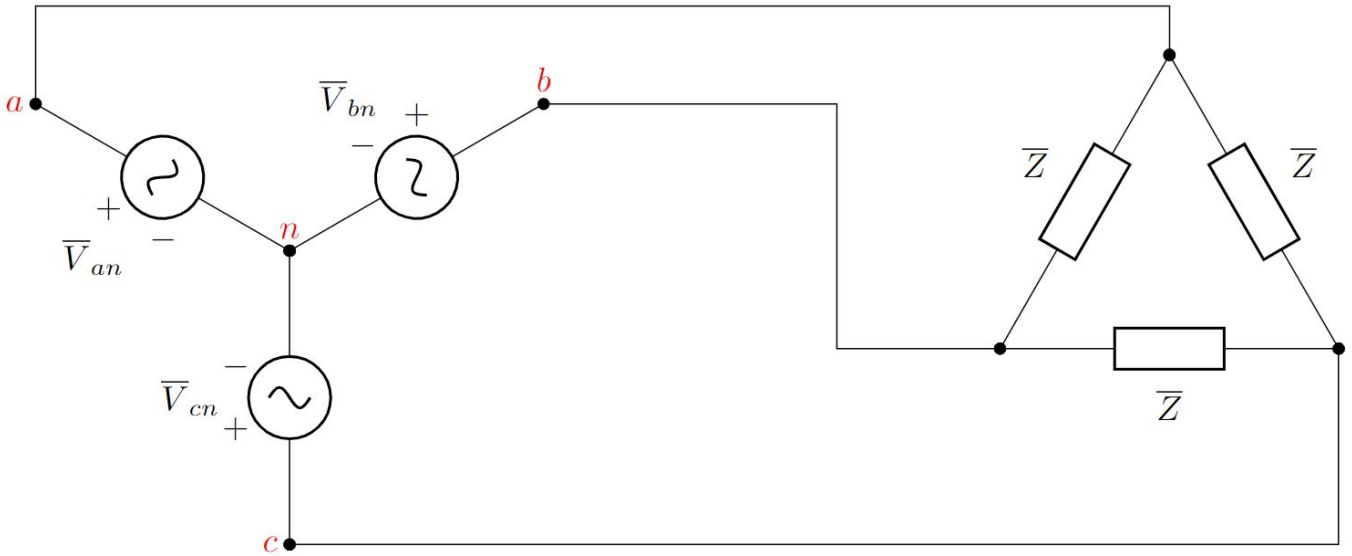
Per phase equivalent: wye-wye connection

$$V_L = \sqrt{3}V_\varphi$$

$$\bar{S}_{3\varphi} = \sqrt{3}V_L I_L \angle\theta$$

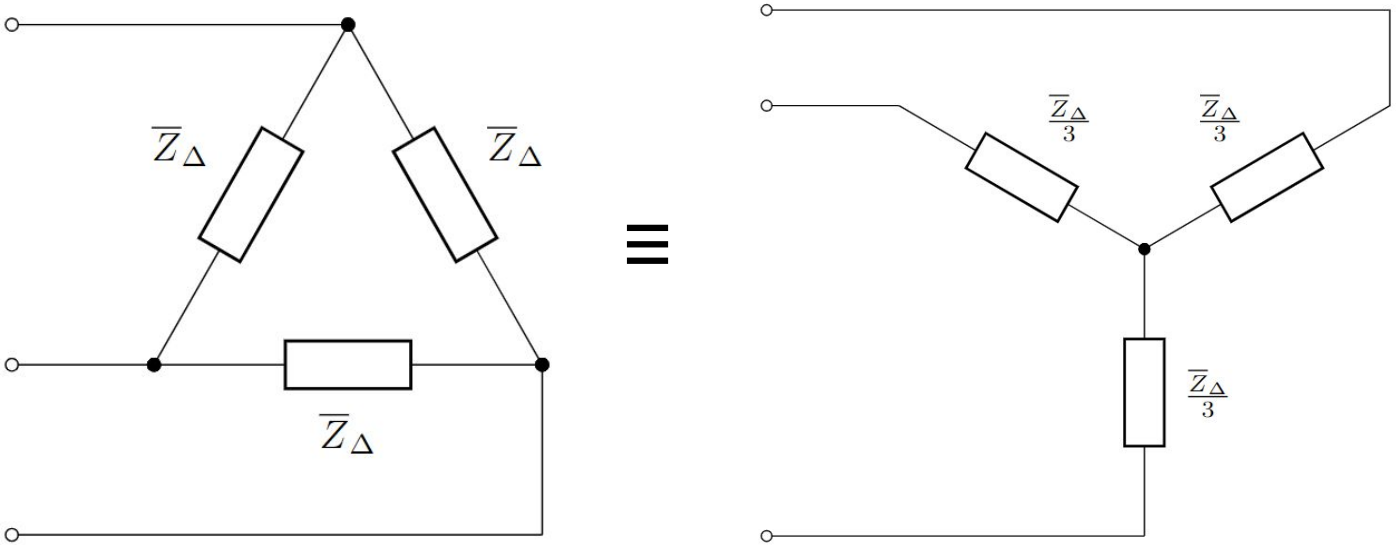
Wye-Delta Connection

- Delta connection: line to line voltage is the same as phase voltage
- Line current and phase current are related by $I_L = \sqrt{3} I_\varphi$
- No neutral connection, can we still do per phase analysis?



Delta-Wye Conversion

- For a balanced load, delta and wye impedances are related by $\bar{Z}_Y = \bar{Z}_\Delta/3$
- To analyze delta-connected load, first convert to wye, then do per phase analysis like before



Example: Spring 2019 Exam 1

Problem 2 (25 points)

Three 3-phase loads are connected in parallel to a wye-connected source with 480 V line to line.
The loads are given as:

1) A wye connected load consuming 250 kVA at a 0.707 lagging power factor
2) A wye connected load consuming 75 kW at a 0.9 lagging power factor
3) A delta connected load with impedance Z=100+j100 Ω

a) What is the total line current supplied by the source? (13 points)

b) A delta connected capacitor bank is added in parallel to the three loads to make the overall power factor 0.99 lagging. What is the new line current? (12 points)

Potentially useful formulas:

sin(x)=cos(x-90°)

0<θ<180° (lag)

-180°<θ<0 (lead)

$\bar{V} = \bar{Z}\bar{I}$

$I_L = \sqrt{3} I_\varphi$ (delta)

$V_L = \sqrt{3} V_\varphi$ (wye)

$\bar{S} = \bar{V}\bar{I}^* = P + jQ$

$\bar{Z}_Y = \bar{Z}_\Delta / 3$

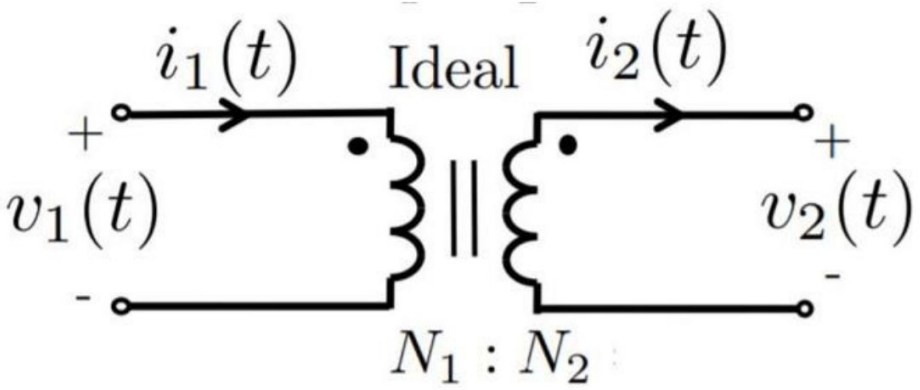
$\mu_0 = 4\pi \times 10^{-7}$H/m

$\bar{S}_{3\varphi} = \sqrt{3} V_L I_L \angle\theta$

Transformers

Ideal Transformer

- No power losses
- Voltage on secondary terminal is scaled by ***a***
- Relations only valid for AC current

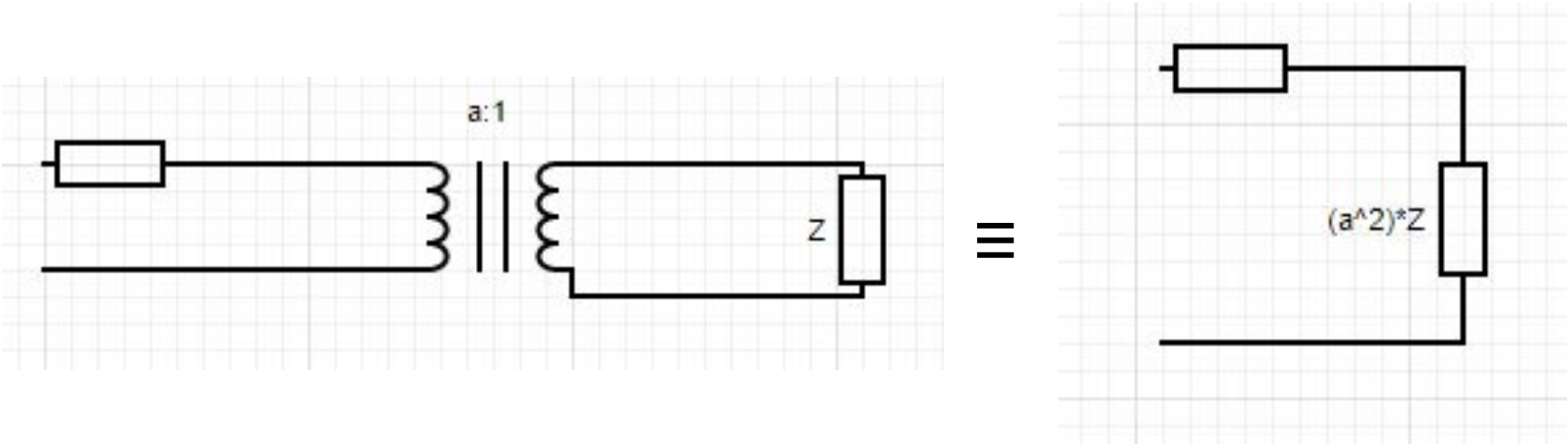


$$N_1 i_1 = N_2 i_2$$

$$a = \frac{N_1}{N_2}$$

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

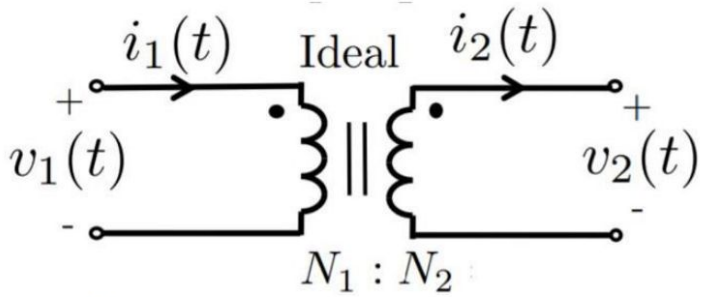
Ideal Transformer - Equivalent Impedance



Example: Spring 2015 Exam 1, #4

b) An ideal transformer has a winding ratio of 2000:200. The primary (high voltage) side is connected to 120V. The secondary side serves a load with impedance 1+j0.5 Ω.
   i. Find the current on the primary winding.
   ii. What does the load impedance look like on the primary side?

Potentially useful formulas:



$$a = \frac{N_1}{N_2} \qquad N_1 i_1 = N_2 i_2$$

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

Magnetic Circuits

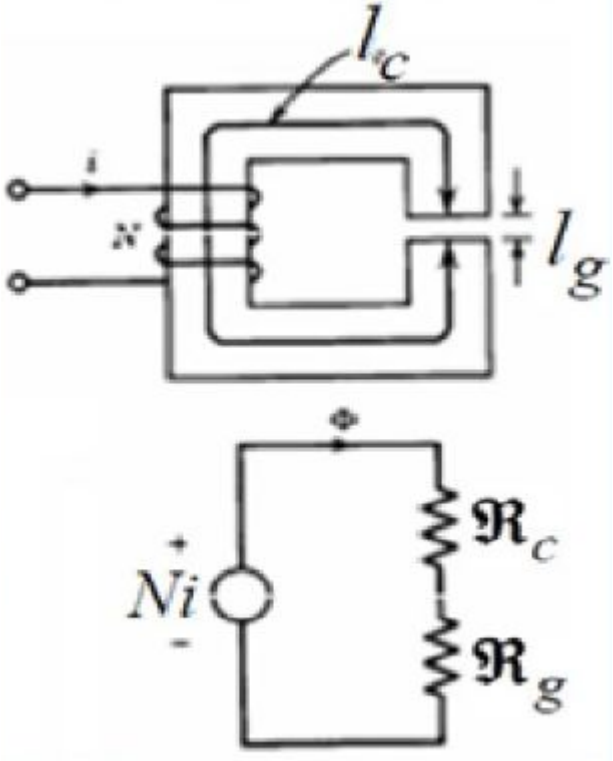
Magnetic Circuits

- Magnetomotive force (mmf)
  - Drives magnetic flux through a magnetic circuit
  - Mmf = Ni
- Magnetic flux

$$\varphi = BA$$

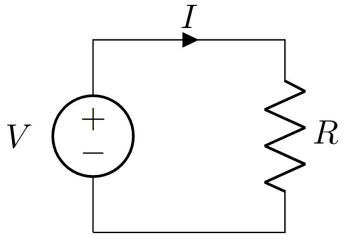
- Reluctance
  - Ratio of mmf to magnetic flux

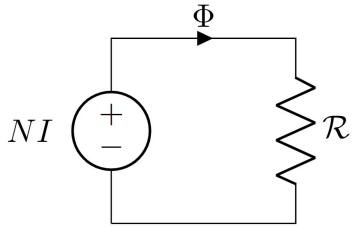
$$\mathcal{R} = \frac{l}{\mu A} \qquad Ni = \mathcal{R}\varphi$$



Electric and Magnetic Circuit Analogs

Electric	Magnetic
Electromotive force (emf) [V]	Magnetomotive force (mmf) [A • turns]
Current [A]	Magnetic flux [Wb]
Resistance [Ω]	Reluctance [A • turns / Wb]





Flux Linkage

- The total magnetic flux linked by N turns of a coil:

$$\lambda = N\varphi$$

- Voltage across the coil is the rate of change of flux linkage:

$$v = \frac{d\lambda}{dt}$$

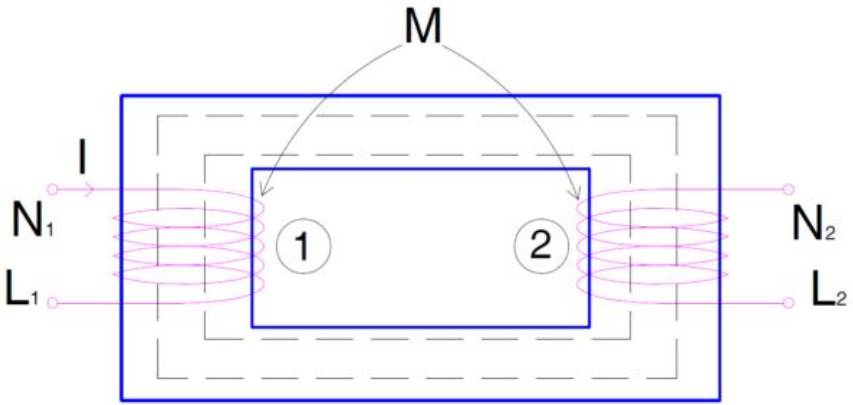
- Voltage across an inductor:

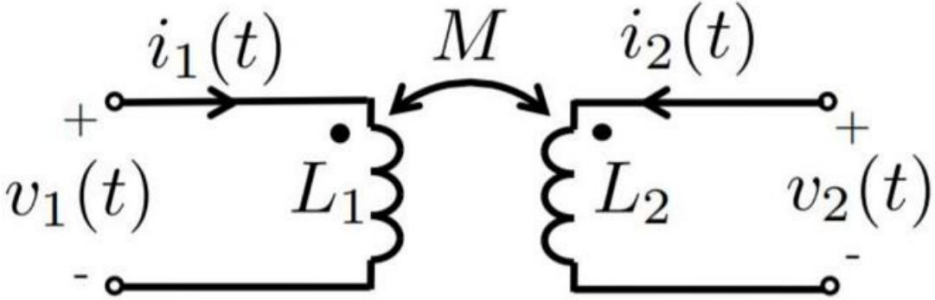
$$v = L\frac{di}{dt}$$

- If system is linear:

$$\lambda = Li$$

Coupled Coils





- Coupling coefficient:

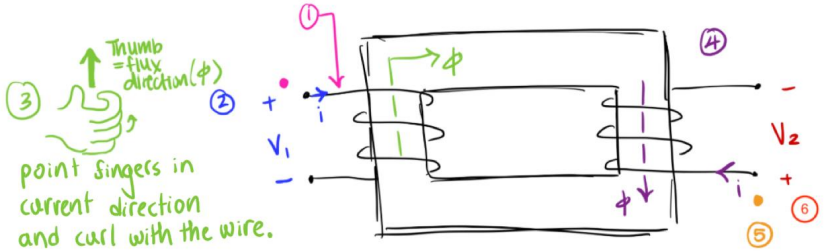
$$k = \frac{M}{\sqrt{L_1 L_2}}$$

$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

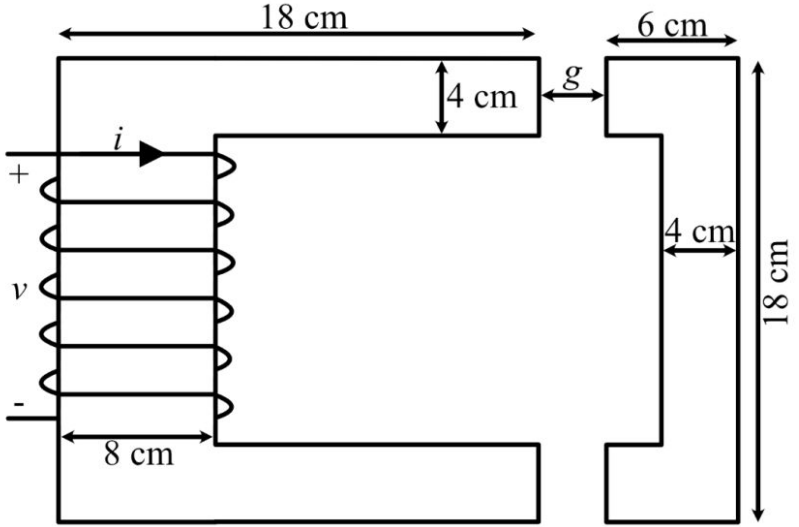
Dot Convention

1) Select a coil and put a dot on one terminal arbitrarily
2) Mark dotted terminal with + voltage polarity and inject current into first dot
3) Use RHR to find flux direction
4) Determine current direction in second winding by making sure the flux through coils are traveling in the same direction
5) Put dot in second winding where current enters terminal
6) Put + voltage polarity on second terminal (current leaves + end of voltage)



Example: Spring 2021 Exam 1

Problem 3 (25 Points)



An iron core with a depth into the page of 5 cm and permeability μ=2500μ_0 has a coil with 250 turns wrapped around it as shown. If the air gap g=10 mm, what is the inductance of the coil? Neglect fringing effects.

Potentially useful formulas:

$$\int \underline{H} \cdot \underline{dl} = \int \underline{J_f} \cdot \hat{n} dA \qquad \int \underline{E} \cdot \underline{dl} = -\frac{d}{dt}\left(\int \underline{B} \cdot \hat{n} dA\right)$$

$$\varphi = BA \qquad \lambda = N\varphi = Li \text{ (if linear)}$$

$$\mathcal{R} = \frac{l}{\mu A} \qquad Ni = \mathcal{R}\varphi$$

$$v = \frac{d\lambda}{dt} \qquad k = \frac{M}{\sqrt{L_1 L_2}}$$

Questions?



Thanks for attending!

Please fill out the feedback form!

